

Lesson 6: Realized Volatility and the Volatility Risk Premium

Analyst onboarding · Lesson 6 of 10 · L/S Gamma Desk, QUANTT

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Goal: measure how much SPY actually moves, and understand the gap between that and what options charge, the edge our desk exists to trade.

You need: Lessons 1, 3 and 5 (volatility, implied volatility, RV vs IV).

1 Measuring realized volatility (RV)

Realized volatility is how much SPY **actually** moved. There are several ways to measure it.

1. Close-to-close. Take daily log returns $r_t = \ln(P_t/P_{t-1})$ over a window, e.g. the last 21 trading days (one month), and compute

$$\sigma_{RV} = \sqrt{252} \times \text{std}(r_t)$$

It's simple and captures overnight gaps, but it's noisy because it uses only one price per day.

2. Range-based (Garman-Klass). Also uses each day's **open, high, low and close**. The high-low range shows how far SPY travelled *during* the day, which gives a sharper estimate from fewer days:

$$\sigma_{GK}^2 = \frac{1}{2} \left(\ln \frac{H}{L} \right)^2 - (2 \ln 2 - 1) \left(\ln \frac{C}{O} \right)^2$$

Our pipeline computes this (`rv_gk` in `pipeline/features.py`). **Caveat:** it ignores the overnight jump from yesterday's close to today's open, so it reads a little **low** compared with VIX.

3. Intraday (high-frequency) RV. Sum the squared 5-minute returns through the day. This is the most accurate, but needs intraday data that our free live pipeline doesn't have.

2 The volatility risk premium (VRP)

Compare what options **charged** (IV) with what then **happened** (RV over the same period):

$$\text{VRP} = \text{IV} - \text{RV}$$

Fact: IV is **usually higher** than the RV that follows.

- From 1990 to 2018, VIX averaged about **19.3%**, while the S&P 500's realized volatility averaged about **15.1%** (Bondarenko, Cboe 2019).
- IV is above subsequent RV about **70–80% of the time**.
- Delta-hedged index options **lose money on average** for buyers, and lose the most at the money (Bakshi & Kapadia 2003).

⇒ On average, **selling options and delta hedging earns the VRP**. That is the short-gamma trade from Lesson 5.

3 Why does the VRP exist?

If options are usually overpriced, why doesn't everyone sell them until the premium disappears?

1. **Insurance demand.** Big investors buy index puts to protect portfolios. Like any insurer, the seller expects a profit on average.
2. **Crash risk.** Short-gamma losses are rare but huge, and arrive when everything else is crashing too. Sellers demand payment for that risk.
3. **The leverage effect.** Volatility spikes when markets fall (Lesson 1), so puts become most valuable exactly when investors are poorest. That is also why the **put skew** exists (Lesson 3).

So the VRP is **compensation for risk**, not a free lunch. In 2008 and March 2020, realized volatility blew far past implied, and short-gamma traders took large losses.

4 Why we don't just always sell

"Always sell options" earns the VRP most of the time, but:

- **It loses big in crashes.** The Cboe PutWrite index, which sells puts every month, had a -32.7% maximum drawdown and a 40-month longest drawdown (1986–2018).
- **The VRP changes over time.** Sometimes IV is cheap relative to the coming moves, and then **buying** options pays.

$\Rightarrow$ Our edge is not "sell vol". It is **forecasting RV better than IV does**, so we know when to sell, when to buy, and when to sit out.

5 How our code uses the VRP

We use the **standard definition**, the same one as every paper you'll read, but with **our forecast** in place of the realized volatility we can't see yet:

$$\text{VRP} = \text{IV} - \text{RV}_{\text{forecast}}$$

- **Positive VRP:** options charge more movement than we expect, so they look **expensive**: sell (short gamma).
- **Negative VRP:** options charge less than we expect, so they look **cheap**: buy (long gamma).

6 The leverage effect in numbers

Volatility and SPY returns are **negatively correlated**: on big down days, IV usually jumps. Two consequences for us:

1. A **short put** loses twice in a selloff: SPY falls (gamma losses) *and* IV rises (vega losses).
2. A **long call** that has been delta hedged gains from the move but can lose vega if IV falls in a rally.

7 In our code

- `pipeline/features.py` builds `rv_gk` and its daily, weekly and monthly averages (`rv_d`, `rv_w`, `rv_m`).
- The live signal logs `rv_forecast`, `iv` and `vrp` every day to `results/paper/signals.parquet`.
- On 2026-10-02: forecast 12.9%, IV 13.0%, so the VRP was +0.001, about zero, and **no trade**.

8 Words to know

- **Realized volatility (RV):** volatility measured from actual price moves.
- **Garman-Klass:** an RV estimate using open, high, low and close.
- **VRP:** IV minus subsequent RV, usually positive.
- **Drawdown:** fall from a previous peak.
- **VRP (our code):** $IV - \text{forecast RV}$.

9 Check yourself

1. Over the next month SPY's realized volatility is 11%, and you sold options at 14% IV. Did you likely make or lose money (before costs)?
2. Why does Garman-Klass RV read lower than VIX on average, even beyond the VRP?
3. Our VRP today is +0.03. Are options cheap or expensive by our forecast?

Answers: (1) Likely made money: $IV > RV$. (2) It misses overnight gaps. (3) Expensive: forecast RV is 3 points below IV.